

Anthony Nelson

Full-Stack Software Engineer (Backend & Infrastructure Focus)

Backend Systems • DNS • Docker • Redis • Platform Engineering

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[GitHub](#) • [LinkedIn](#) • [Portfolio](#)

Summary

Backend Software Engineer with hands-on experience building **DNS management platforms**, **containerized backend systems**, and scalable **REST APIs**. Experienced with Node.js, Express.js, MongoDB, Redis, Docker, Nginx, BIND9, Linux, CI/CD deployment, and asynchronous job processing.

Technical Skills

Languages: JavaScript, TypeScript, Python

Backend & Databases: Node.js, Express.js, MongoDB, SQL, Redis

Infrastructure, DevOps & Systems: Linux, Docker, Nginx, BIND9, DNS, BullMQ, System Design (Foundations)

Frontend: React.js, HTML, CSS, Tailwind CSS

Tools: Git, GitHub, Postman, VS Code

Professional Experience

Ocellus Business & Services – Masjid, Mumbai

Software Developer | April 2026 – Present

- Designed, developed, **deployed**, and **maintained** the company's primary **MERN platform** as the **sole software developer**, including an **admin panel** with 20+ features.
- Developed an automated **invoice generation** system featuring a live preview interface, utilizing **Puppeteer** to dynamically generate **downloadable PDF documents** from live application data.
- Set up a secure on-premise **Linux server** using **Cloudflare Tunnel**, removing the need to expose inbound ports.
- Established an automated **CI/CD deployment pipeline** using **GitHub Actions**, **Docker Hub**, and **Watchtower**.
- Implemented and automated the backup process for production databases using **Rclone** to ensure resilience in the **disaster recovery** strategy.

Website: www.ocellus.co.in

Projects

DnsTitle | DNS Management & Subdomain Hosting Platform

- Built a **DNS management platform** enabling automated **domain verification**, **subdomain provisioning**, and **dynamic DNS record management**.
- Implemented **asynchronous job processing (Redis + BullMQ)** to reliably handle DNS updates and domain lifecycle workflows.
- Integrated **BIND9** to support **dynamic DNS record creation, updates, and automated propagation**.
- Configured **Nginx reverse proxy** to enable **scalable multi-tenant routing and domain hosting**.
- Implemented **rate limiting and basic validation** to prevent abuse and ensure API stability.
- Built a **React + TypeScript dashboard** for managing domains, subdomains, and DNS records with API-driven UI updates.
- Used **Redux Toolkit** for global state management (auth, domain data, async job status) and built responsive UI with **Tailwind CSS**.

Tech Stack: React.js, Node.js, Express.js, MongoDB, Redis, BIND9, Docker, Nginx

GitHub: [View on GitHub](#)

Hosting Platform | Containerized PaaS for App Deployment

- Developed a **Platform-as-a-Service (PaaS)** to automate **application deployment and container lifecycle orchestration** using **Dockerode**.
- Designed and implemented **REST APIs** to manage **application deployments**, **container lifecycle operations**, and **deployment status monitoring**.
- Designed **per-user container isolation** to improve **security, performance stability, and resource control**.
- Architected integration with **DnsTitle** to enable **automatic custom domain provisioning and DNS routing** for hosted applications.
- Built a **scalable backend orchestration layer** to manage **builds, deployments, and runtime monitoring**.

Tech Stack: Node.js, Express.js, Docker, Dockerode

GitHub: [View on GitHub](#)

Education

Bachelor of Science in Information Technology

SIES College (Autonomous), Navi Mumbai

Graduation Year: 2026

Achievement: 2nd Place, Inter-College Hackathon